

**K18F 0381**

Reg. No. : .....

Name : .....

**Fourth Semester B.Tech. Degree (Suppl. – Including Part Time)****Examination, May 2018****(2007 Admission Onwards)****PT2K6/2K6 CE/ME/EE/EC/CS/IT/AEI 401 : ENGINEERING MATHEMATICS – III**

Time : 3 Hours

Max. Marks : 100

**Instruction : Answer all questions.****PART – A**

1. a) Show that the function  $u = 3x^2y - y^3$  is harmonic and find a corresponding analytic function  $f(z) = u(x, y) + iv(x, y)$ .
- b) Find the image of the semi-infinite strip  $x > 0, 0 < y < 2$  under the transformation  $w = iz + 1$ . Show the regions graphically.
- c) Using Cauchy's integral formula, evaluate  $\int_c \frac{\cos \pi z^2}{(z-1)(z-2)} dz$  where  $c: |z| = \frac{3}{2}$ .
- d) Expand  $f(z) = \sin z$  in a Taylor's series about  $z = \frac{\pi}{4}$ .
- e) Show that the correlation coefficient is less than or equal to 1 in magnitude.
- f) Define joint probability mass function on two random variables X and Y.
- g) Find a solution of the Laplace equation  $u_{xx} + u_{yy} = 0$  by the method of separation of variables.
- h) Using D' Alembert's method, find the deflection of a vibrating string of unit length having fixed ends with initial velocity zero and initial displacement  $f(x) = a \sin^2(\pi x)$ . **(8×5=40)**

**PART – B**

2. a) Find the bilinear transformation which maps the point  $z = -2i, i, \infty$  onto the points  $w = 0, -3, \frac{1}{3}$  respectively. Find the image of  $|z| < 1$ . **15**

**OR****P.T.O.**



b) 1) Derive the Cauchy-Riemann differential equations of analytic functions. 8

2) Show that  $u = xy$  is harmonic and find a corresponding analytic function  $f(z) = u + iv$ . 7

3. a) 1) Evaluate  $\int_0^{2\pi} \frac{1}{(25 + 7 \cos \theta)} d\theta$  using residues. 8

2) Prove that  $\int_0^{\infty} \frac{x^2}{(x^2 + 1)(x^2 + 4)} dx = \frac{\pi}{6}$ . 7

OR

b) 1) Find Laurent's series in the region  $2 < |z| < 3$  if  $f(z) = \frac{z^2 - 1}{(z + 2)(z + 3)}$ . 8

2) State and prove Cauchy's residue theorem. 7

4. a) For the joint probability distribution of two random variables X and Y is given below.

| X \ Y | Y               |                |                |                | Total           |
|-------|-----------------|----------------|----------------|----------------|-----------------|
|       | 1               | 2              | 3              | 4              |                 |
| 1     | $\frac{4}{36}$  | $\frac{3}{36}$ | $\frac{2}{36}$ | $\frac{1}{36}$ | $\frac{10}{36}$ |
| 2     | $\frac{1}{36}$  | $\frac{3}{36}$ | $\frac{3}{36}$ | $\frac{2}{36}$ | $\frac{9}{36}$  |
| 3     | $\frac{5}{36}$  | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{1}{36}$ | $\frac{8}{36}$  |
| 4     | $\frac{1}{36}$  | $\frac{2}{36}$ | $\frac{1}{36}$ | $\frac{5}{36}$ | $\frac{9}{36}$  |
| Total | $\frac{11}{36}$ | $\frac{9}{36}$ | $\frac{7}{36}$ | $\frac{9}{36}$ | 1               |

Find :

1) Marginal distributions of X and Y.

2) Conditional distribution of X given the value of  $Y = 1$  and that of Y given the value of  $X = 2$ .

OR

(7+8=15)



b) From the following data, find

- 1) The two regression equations
- 2) The coefficient of correlation between the marks in Economics and Statistics.
- 3) The most likely marks in Statistics when marks in Economics are 30.

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|                     |    |    |    |    |    |    |    |    |    |    |
|---------------------|----|----|----|----|----|----|----|----|----|----|
| Marks in Economics  | 25 | 28 | 35 | 32 | 31 | 36 | 29 | 38 | 34 | 32 |
| Marks in Statistics | 43 | 46 | 49 | 41 | 36 | 32 | 31 | 30 | 33 | 39 |

5. a) A string of length  $l$  is initially at rest in equilibrium position and each of its points is given the velocity  $\left(\frac{\partial u}{\partial t}\right)_{t=0} = b \sin^3\left(\frac{\pi x}{l}\right)$ . Find the displacement  $y(x, t)$ .

15

OR

- b) 1) Derive one dimensional heat flow equation.

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- 2) Solve the equation  $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$  with boundary conditions  $u(x, 0) = 3 \sin n\pi x$ ,

$u(0, t) = 0, u(1, t) = 0$ , where  $0 < x < 1, t > 0$ .

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